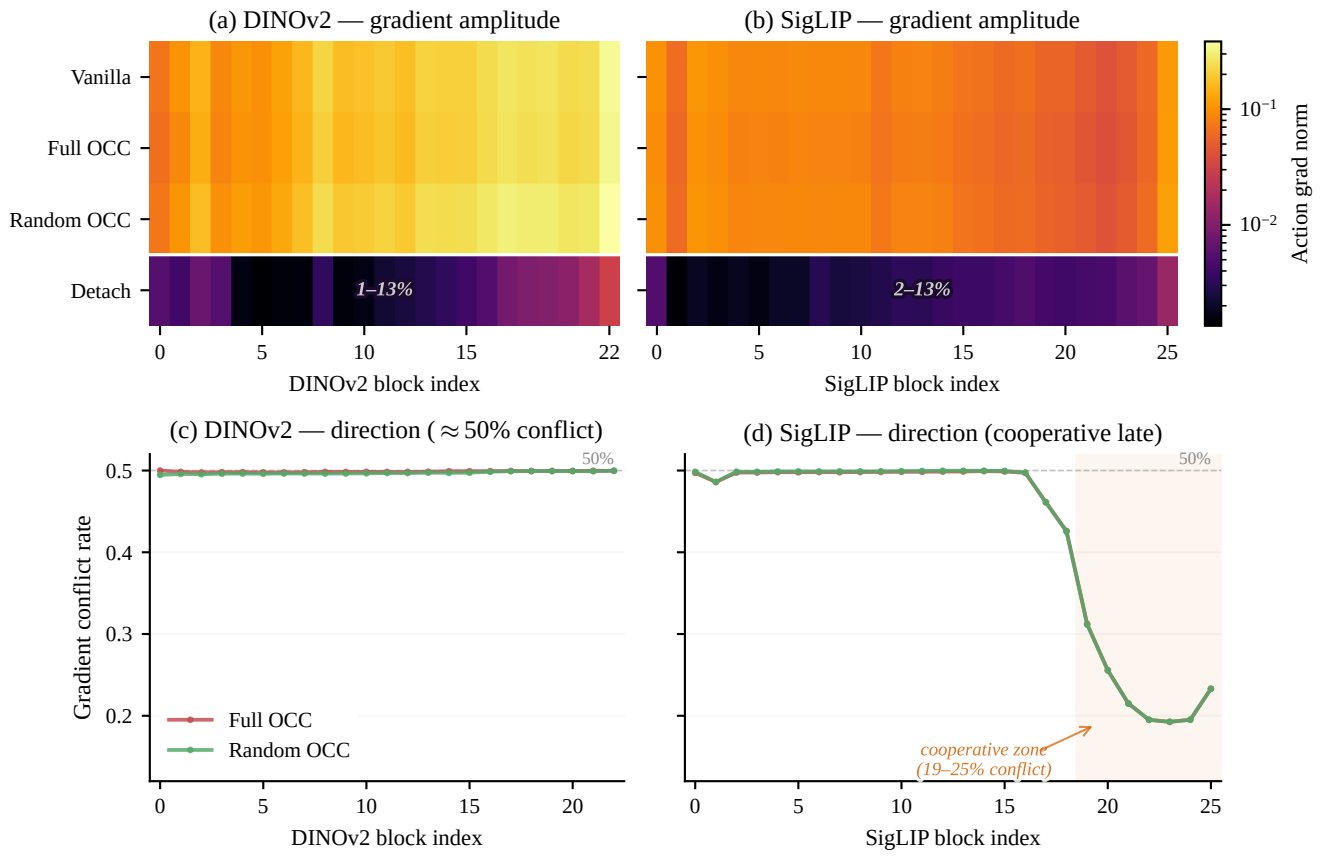
(a) DINOv2 — gradient amplitude
(b) SigLIP — gradient amplitude
Vanilla
Full OCC
Random OCC
Detach
1–13%
2–13%
0
5
10
15
22
20
25
DINOv2 block index
SigLIP block index
10^{-1}
10^{-2}
Action grad norm
(c) DINOv2 — direction (≈50% conflict)
(d) SigLIP — direction (cooperative late)
Gradient conflict rate
0.5
0.4
0.3
0.2
50%
Full OCC
Random OCC
cooperative zone
(19–25% conflict)